PROTOCOL LABS IN VMWARE WORKSTATION 17 PRO

- Ubuntu server VM (192.168.140.150)
- Kali Linux VM (192.168.140.148)
- Win10Pro VM (192.168.140.147)

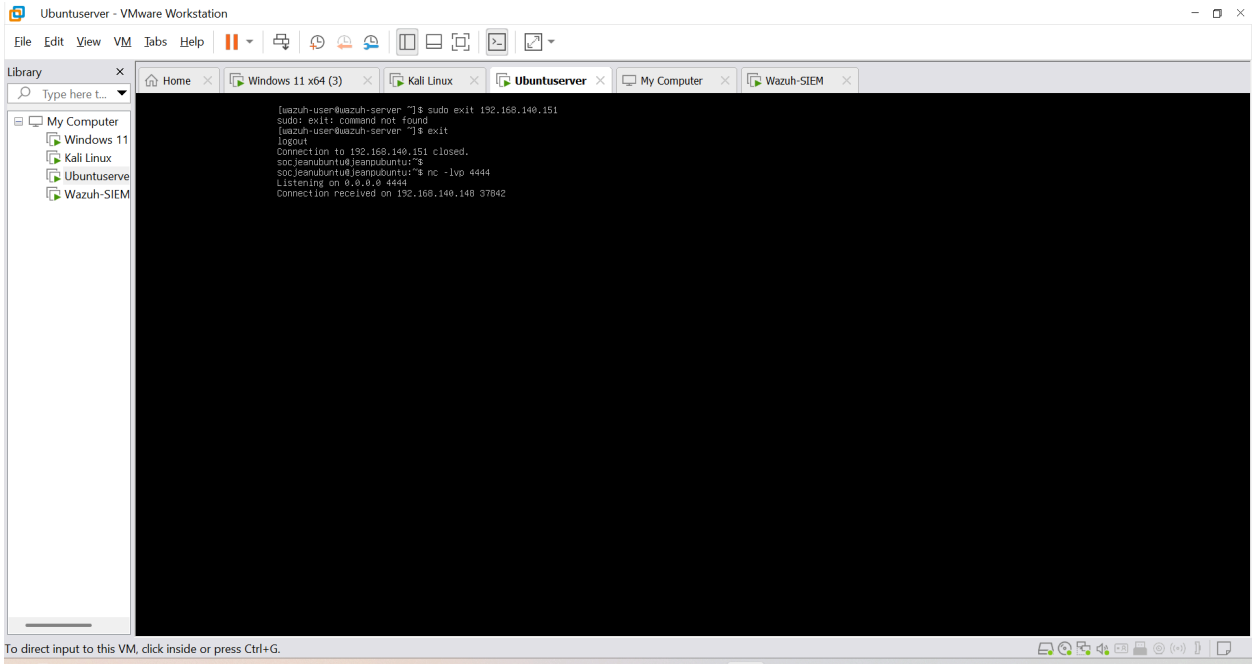
Infrastructure & Lab Architecture

The closed-loop laboratory environment was built using VMware workstations across distinct network subnets to isolate offensive operations from defensive detection:

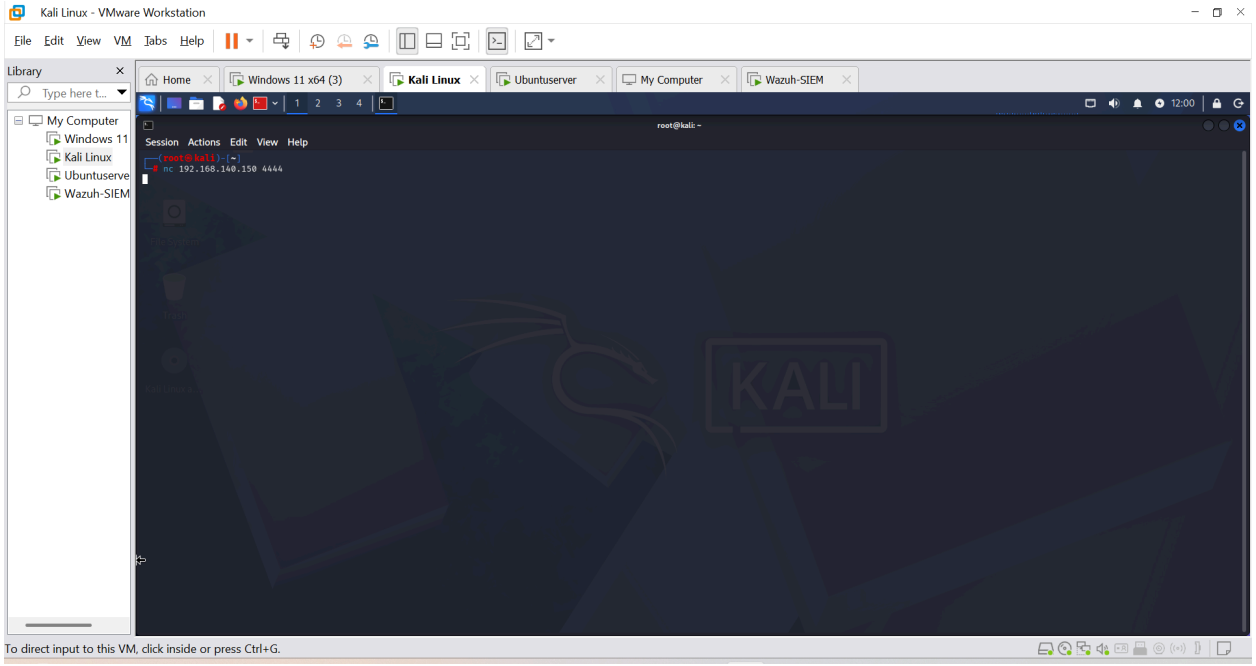
- **Offensive Node:** Kali Linux VM (Attacker machine running custom scripts and payload delivery tools).
- **Target Application Host:** Ubuntu Server VM (192.168.140.150 running an isolated Docker engine).
- **Target Enterprise Endpoint:** Windows 10/11 Pro VM (Configured with System Monitor/Sysmon logging to simulate a corporate user workstation).
- **SIEM / Defensive Analytics Engine:** Wazuh SIEM Server (Centralized log processor, manager, and dashboard indexer).

1. Netcat PROTOCOL

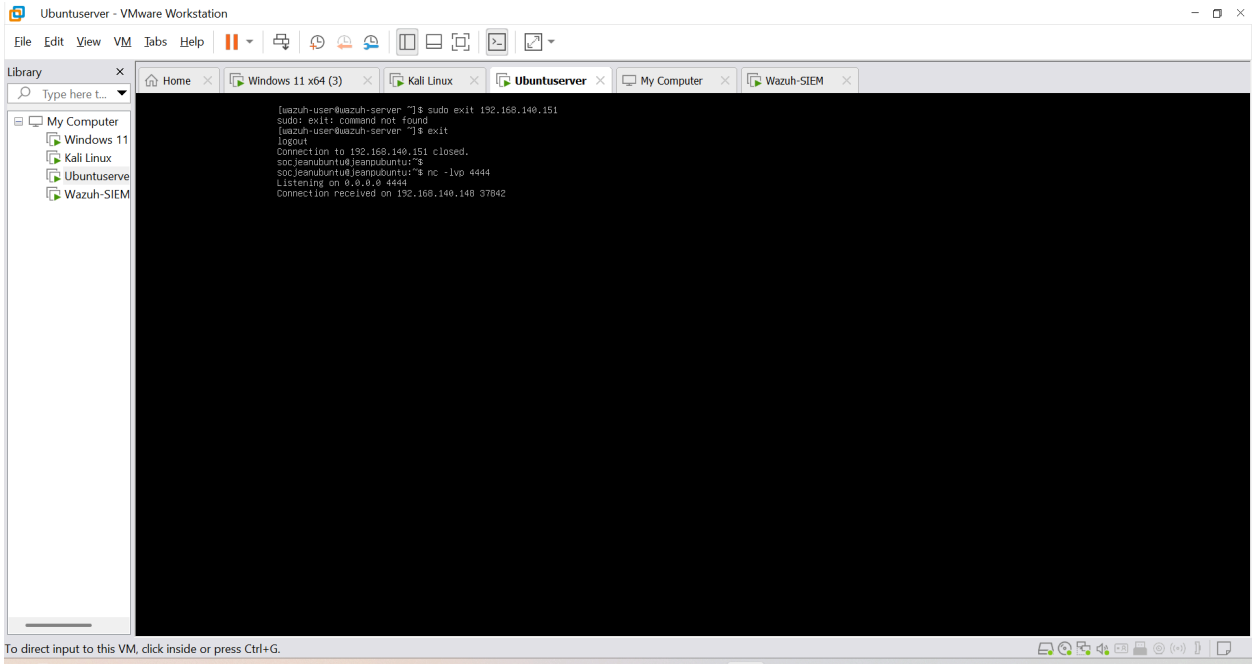
The Netcat command is a versatile networking utility used for reading from and writing to network connections via TCP or UDP. Also called a “Swiss army knife” of networking, it is used for network debugging, testing connectivity, port scanning, and transferring data.
It uses switches depending on the task at hand. Below are some screenshots of the execution of Netcat.



a. Screenshot of Netcat execution from Ubuntu server (192.168.140.150) to connect to Kali Linux VM (192.168.140.148). Network connection initiated with switch combination (`nc -lvp 4444`).

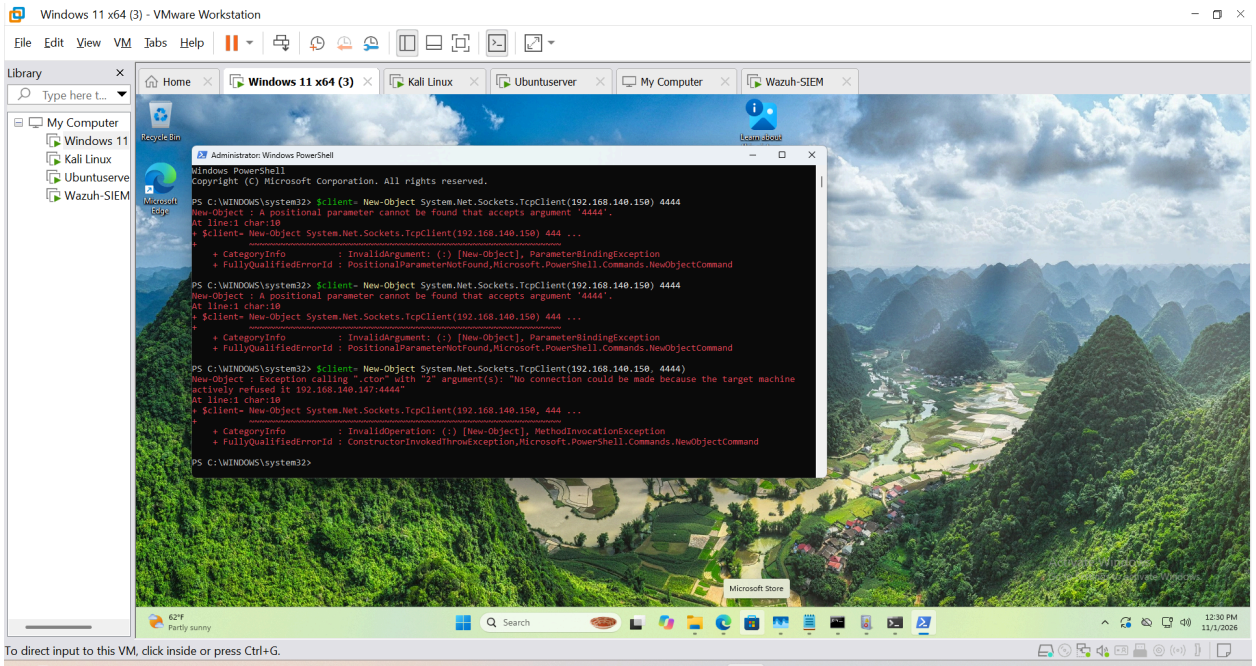


b. Verification of connection established between Ubuntu server and Kali Linux showing on Kali Linux command line interface as "`nc 192.168.140.150 4444`" that can be used to access the server from Kali Linux.

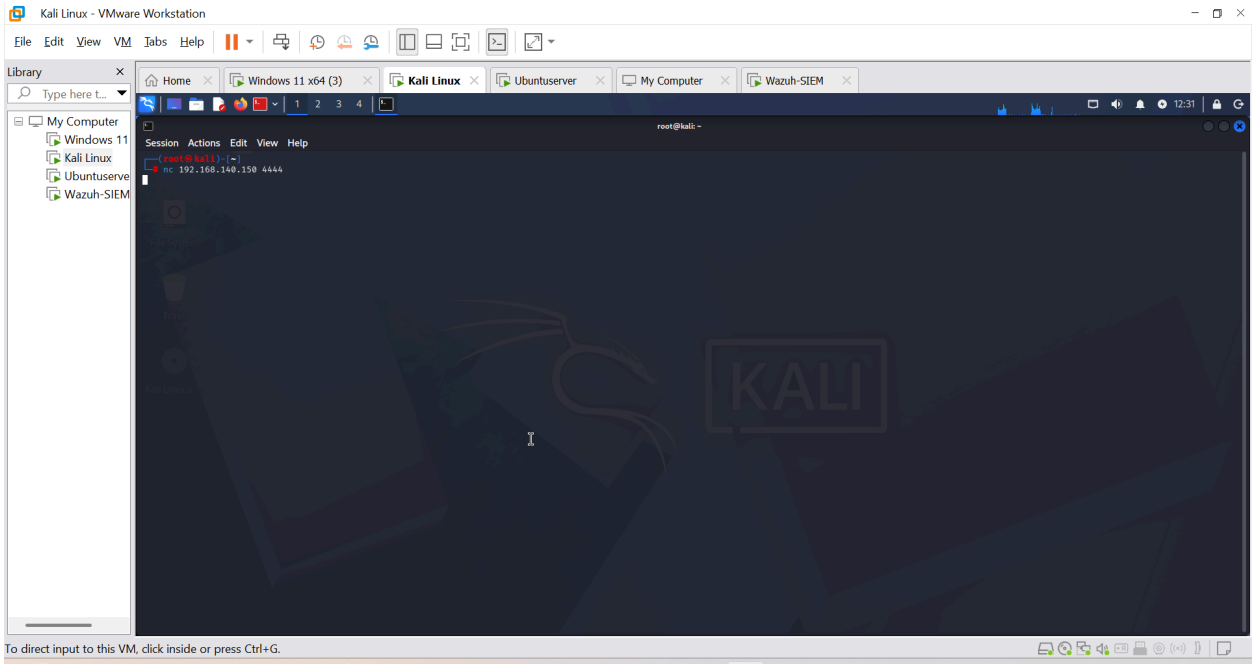


c. This screenshot confirms that the connection has been acknowledged by the server as active and ongoing (received) on port 37842.

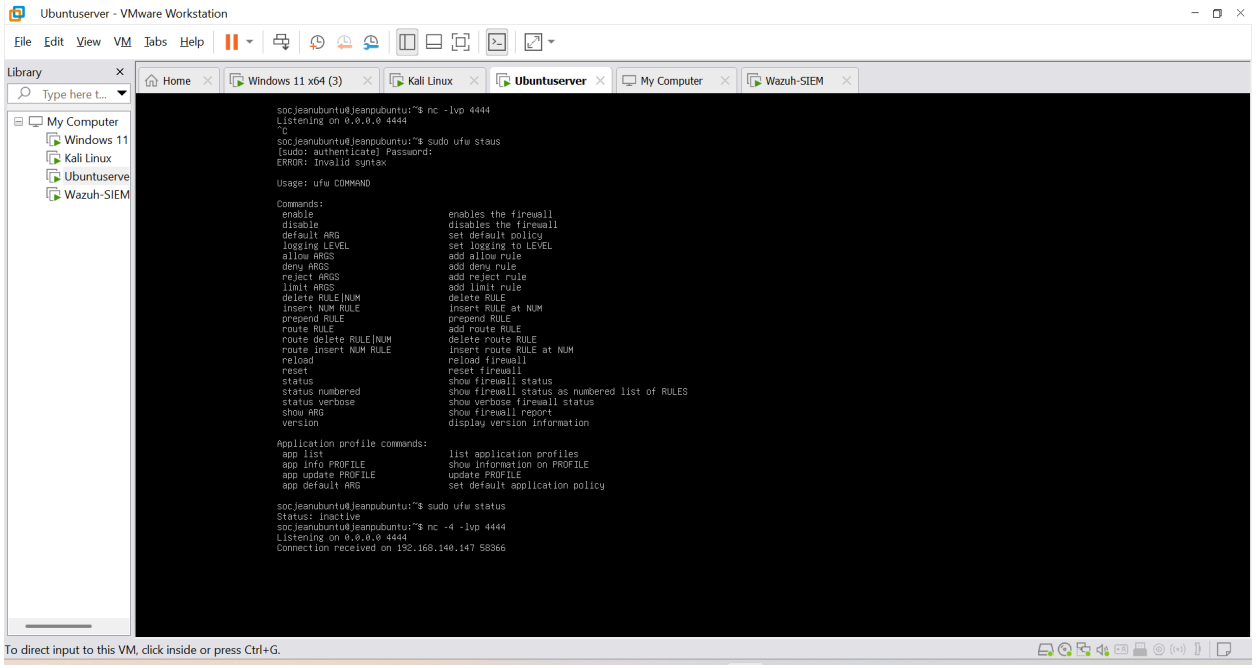
A few attempts to connect to Win10Pro VM failed as shown on the screenshot below and there was a network firewall status assessment performed to verify the reason for the failure due to port 4444 still in listening mode on Kali Linux.



d1. Failed network connection attempts between Ubuntu server and Win10 VMs.

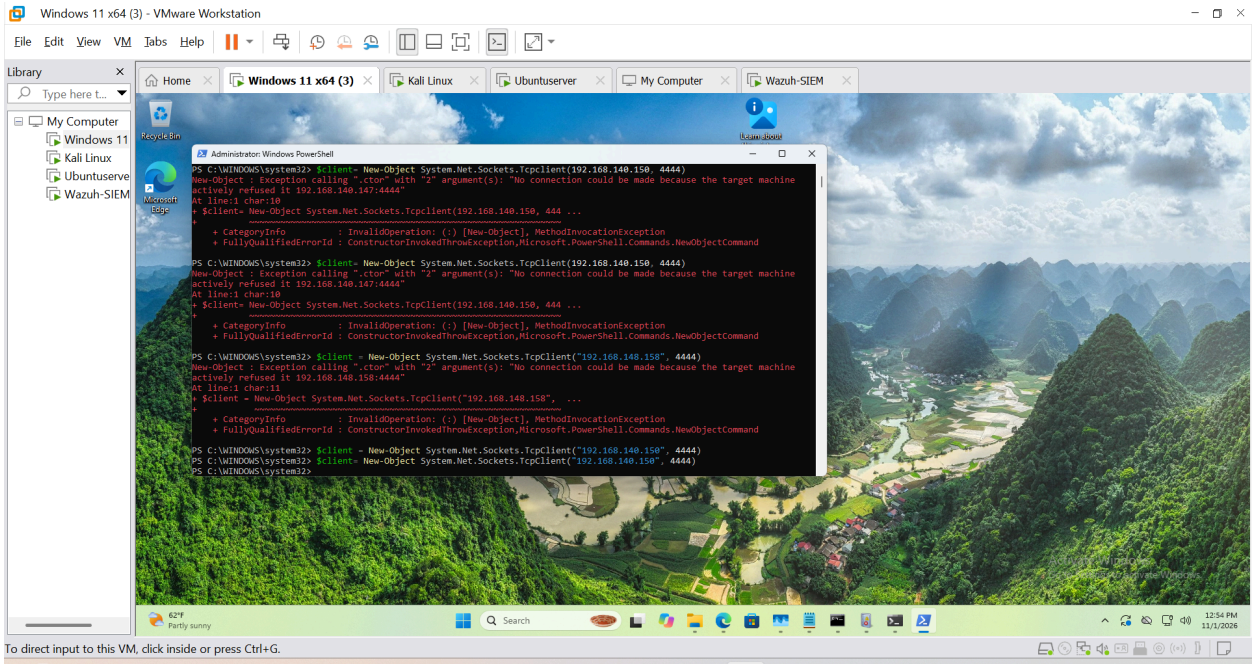


d2. Reason for the connection failure on port 4444 with Ubuntu server.

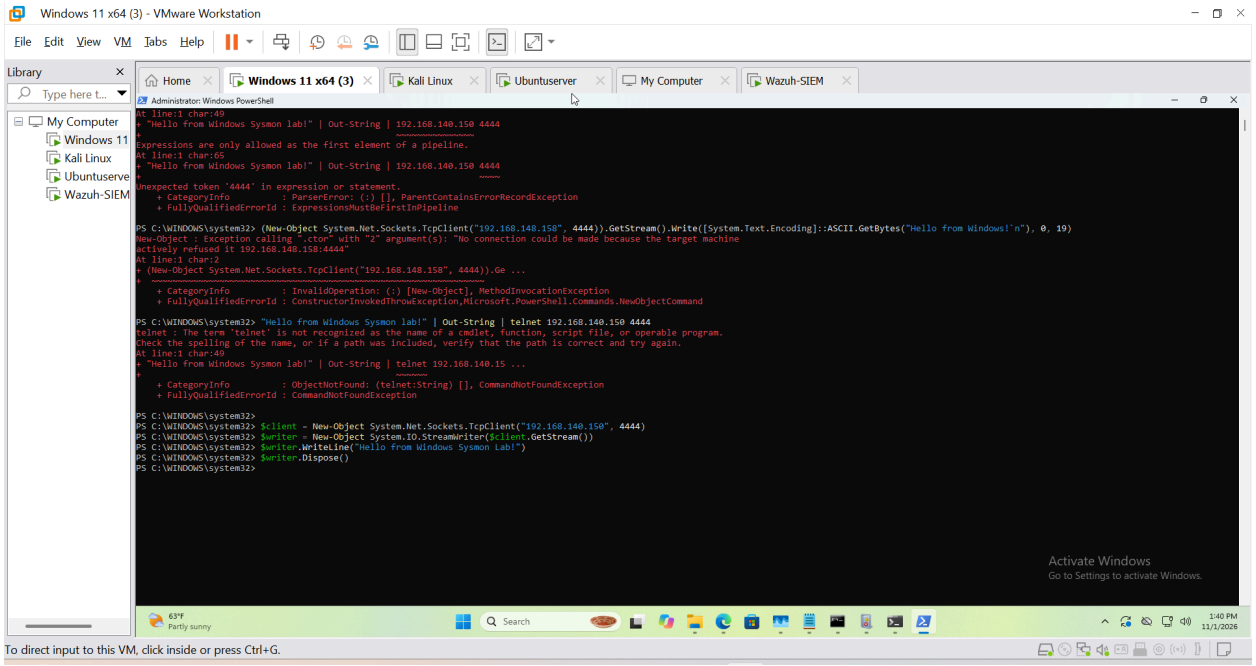


d3. Troubleshooting to assess reasons for failure with firewall status. Then connection was established with Win10Pro VM (192.168.140.147 -4444) after dropping the connection with Kali Linux.

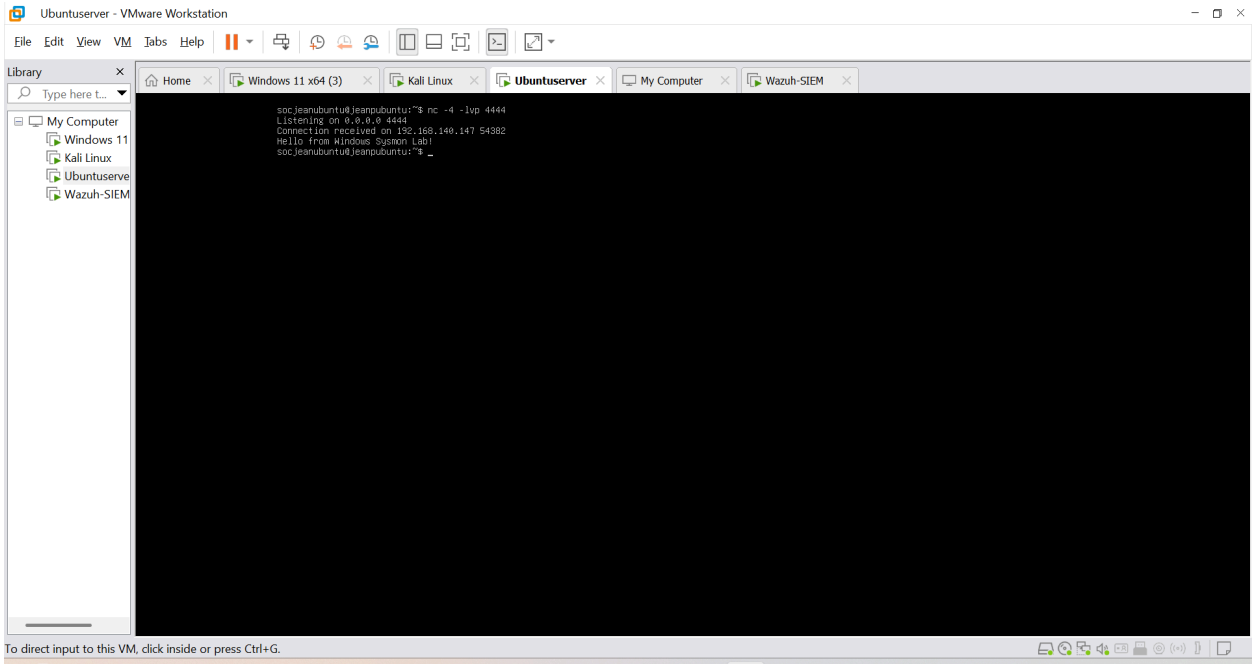
This demonstrates that Netcat can only accept one network connection at a time on a single listening socket.



d4. Netcat connection established with Ubuntu (192.168.140.150 -4444). The system is ready to initiate data transfer from Win10Pro VM (192.168.140.147) to Ubuntu server through TCP. A message is written in PowerShell and the stream will be transferred to Ubuntu server CLI flawlessly as shown below.



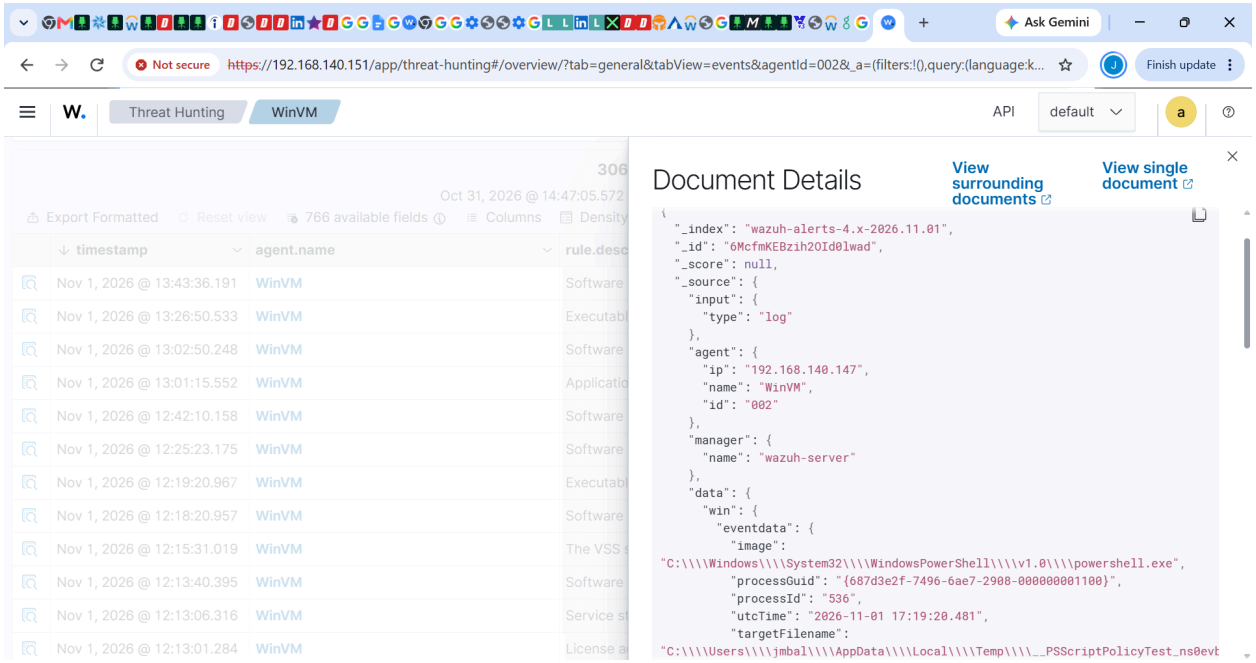
d5. Sender's screenshot from Win10Pro: "Hello from Win10 Sysmon lab!" message.



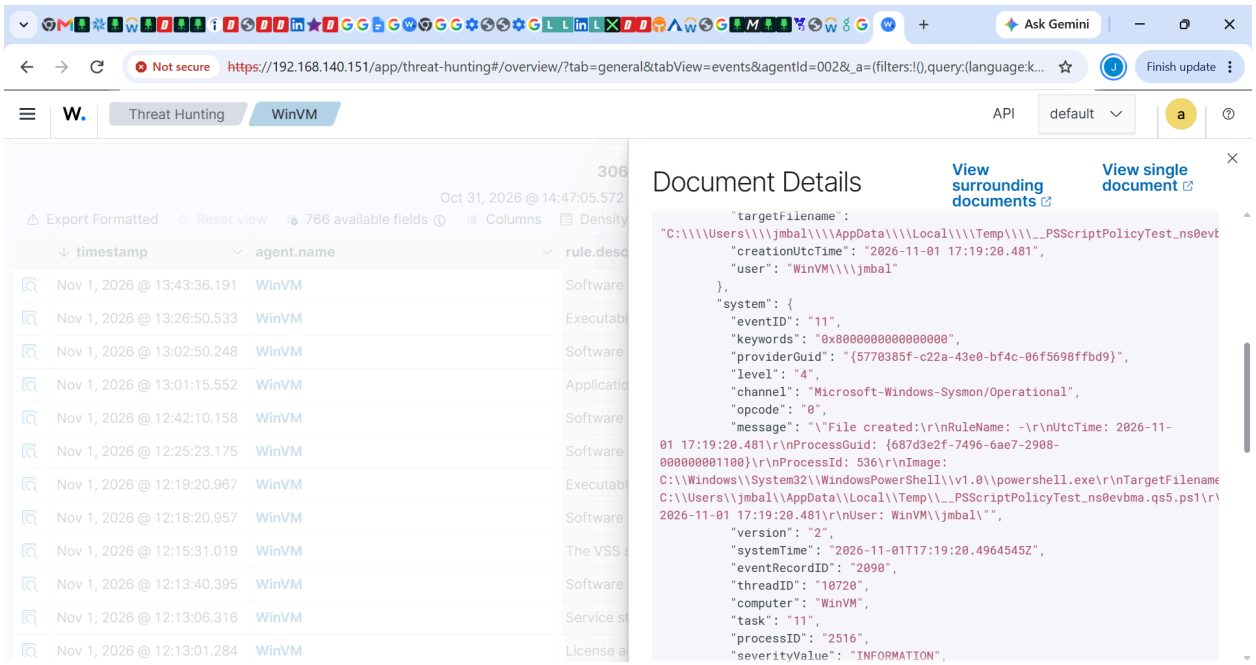
d6. Received message on Ubuntu server showing data exchange after TCP 3-way handshake connection.
This lab demonstrates how Netcat with its switches can be used to connect systems and exchange data as an example of its utility.

2- Verification of Security Logs and Sysmon Logging of Activities

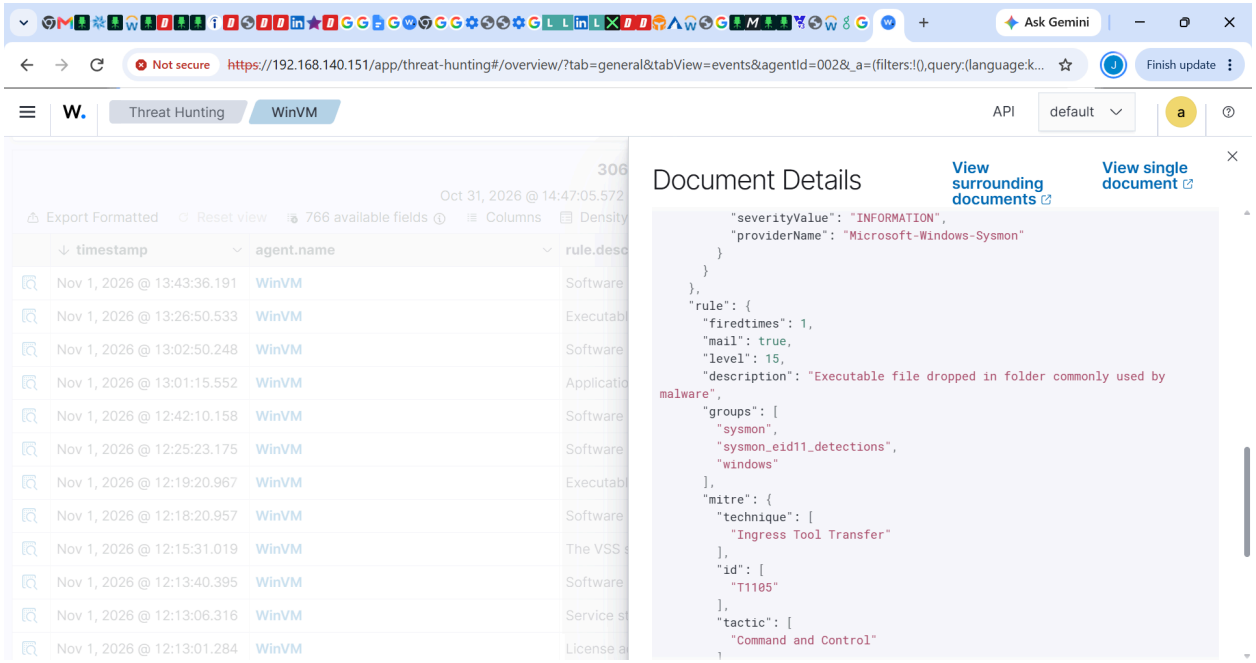
Upon completing initial activities with the Netcat command on a random port 4444 on win10Pro VM using Powershell, one will examine the WAZUH SIEM logs generated from these activities and understand its description and correlation.



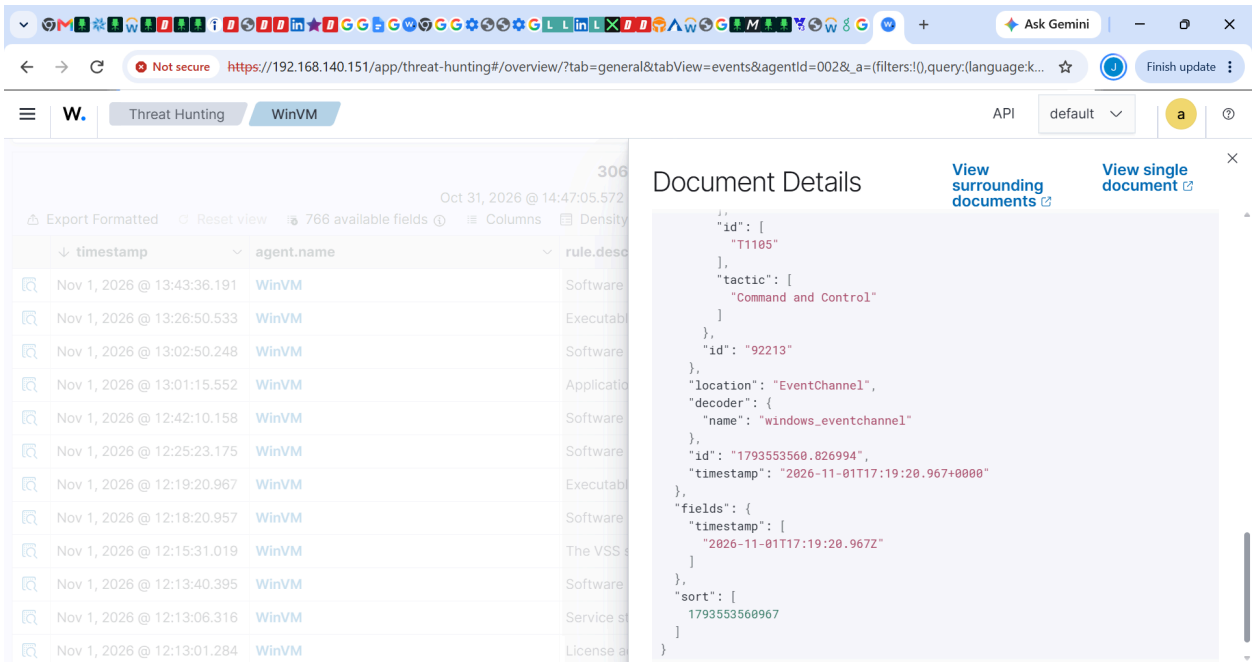
a1-Screenshot displaying the alert top section details in a JSON format in the SIEM tool



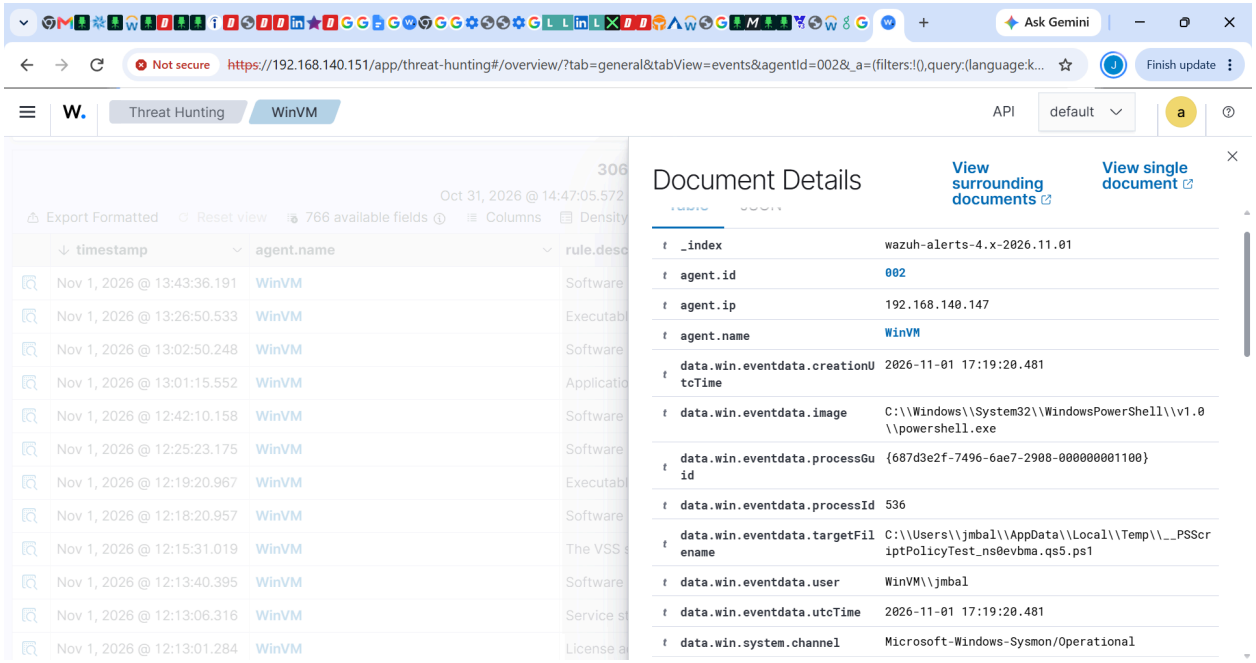
a2-Screenshot of the same alert in JSON format with more information on the action as logged by the SIEM and Sysmon.



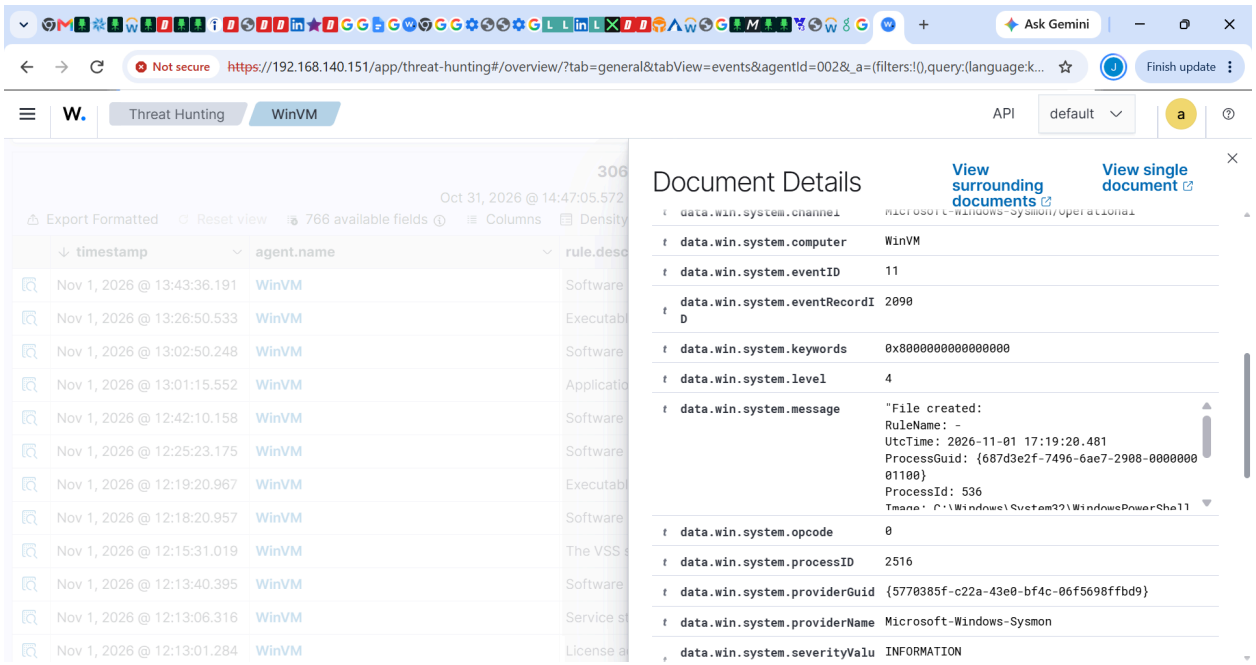
a3-Screenshot of the same alert in JSON with alert level(15),description and alert eid11 detection rule with Mitre technique and T1105 as id then the C2 tactic clearly correlated.



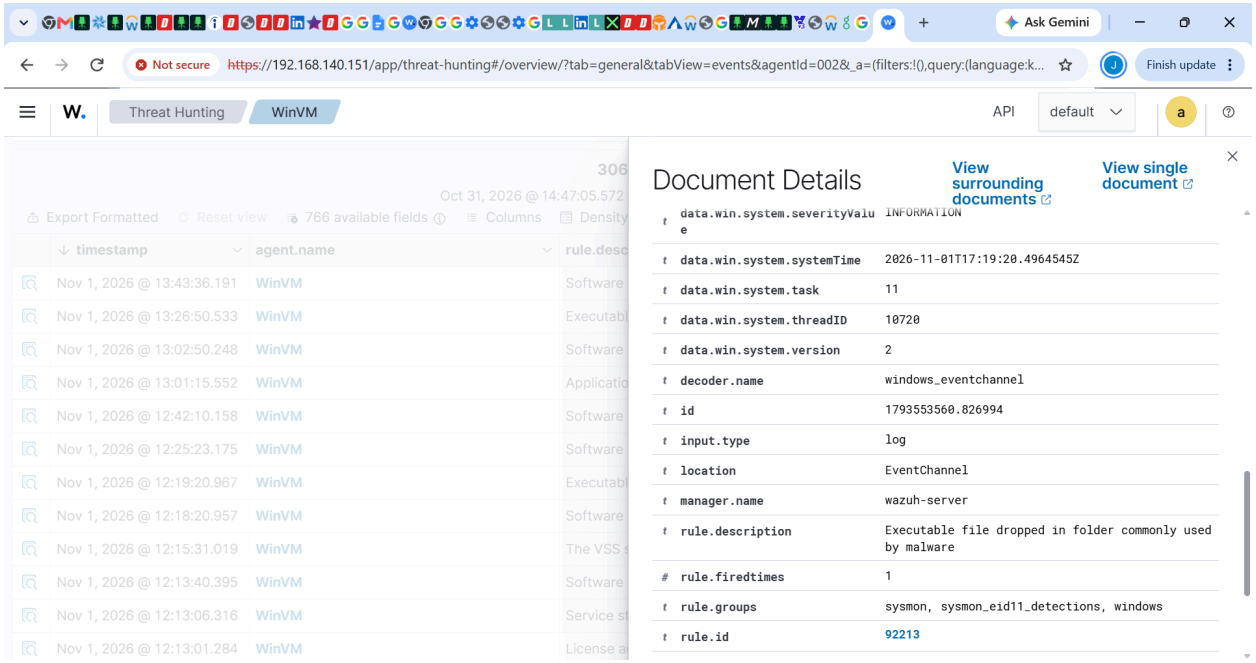
a4-Screenshot showing the same alert id(92213) with timestamp displayed.



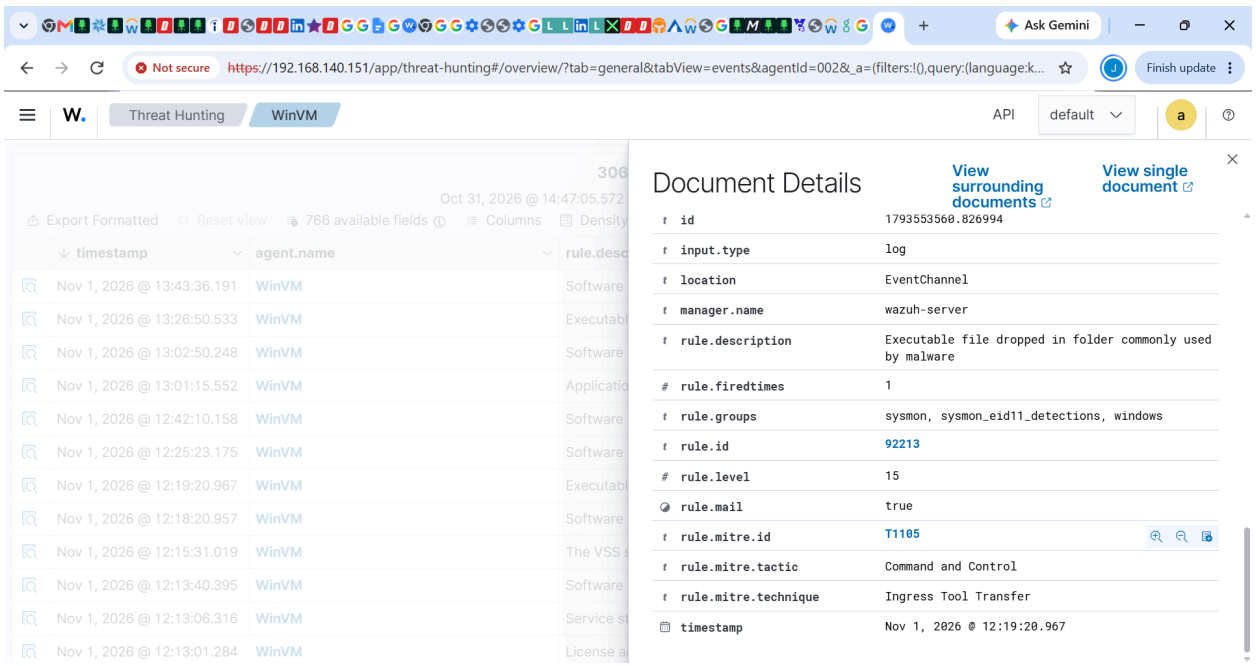
b1-Screenshot of the alert as shown in the top of the details table.



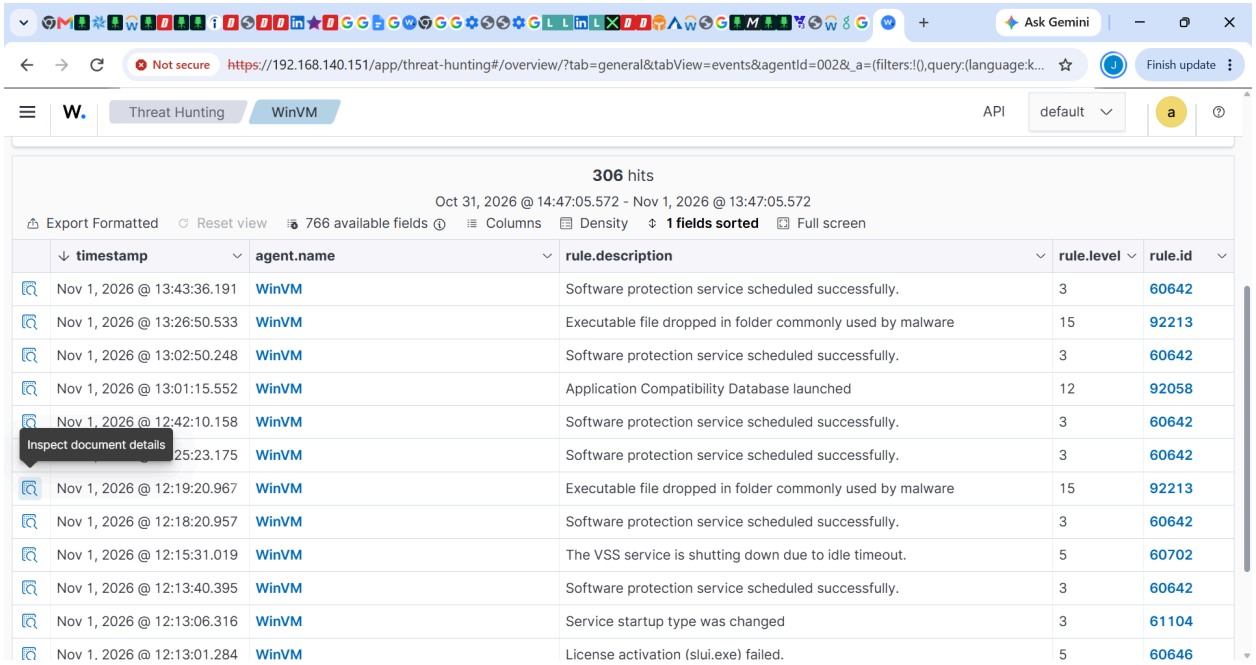
b2-Screenshot of alert section in table event id(11), system level(4), system message(file created), process id(2516), file hash.



b3-Screenshot showing table details with system information,thread id(10720),version(2), input type(log), origin(wazuh-server), rule description(executable…), number of hits(1), and the rule group.

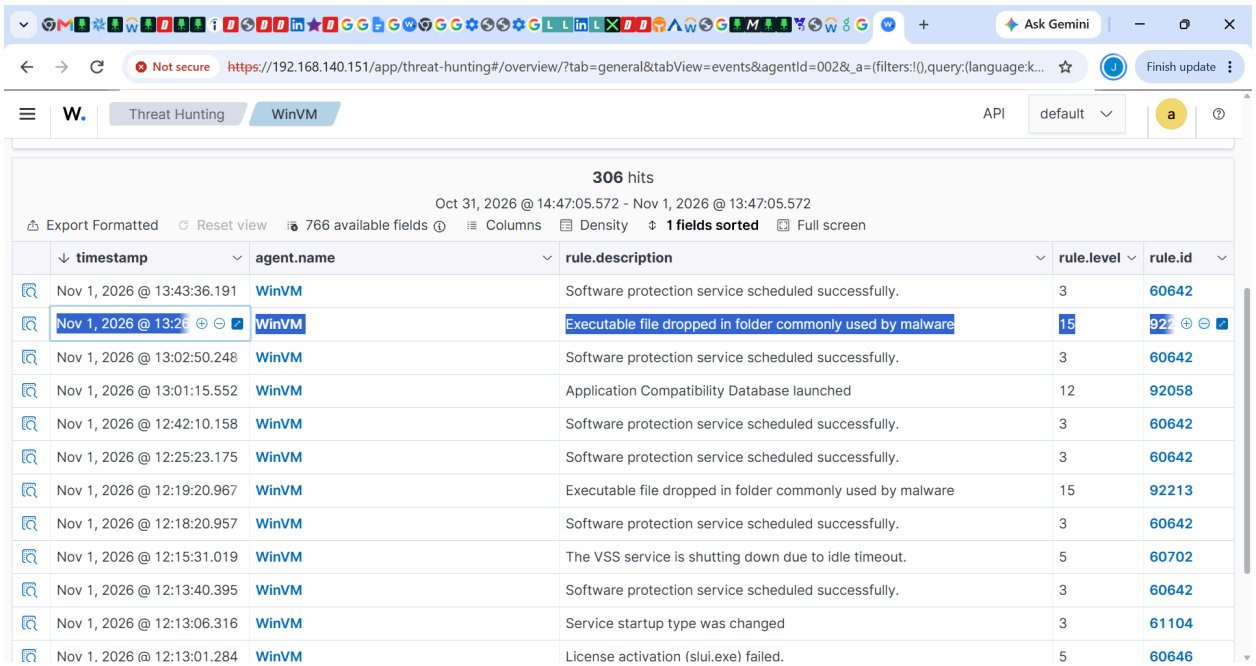


b4-Screenshot of table details with rule level(15), rule mail(true), Mitre(T1105).



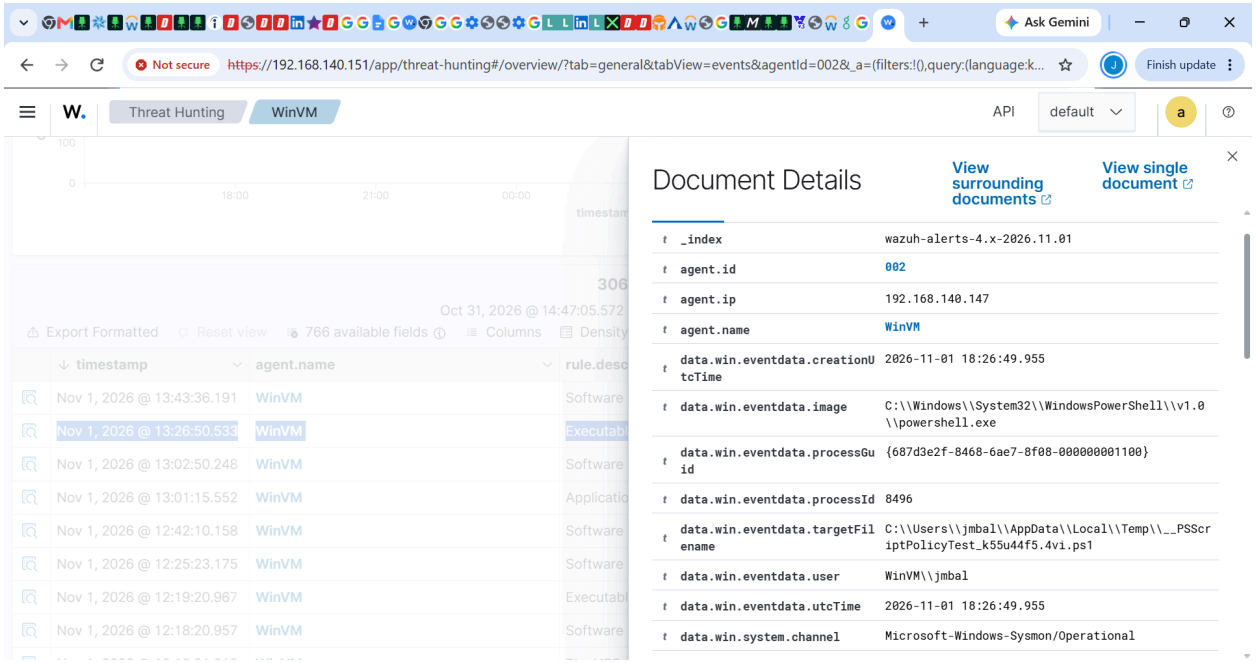
timestamp	agent.name	rule.description	rule.level	rule.id
Nov 1, 2026 @ 13:43:36.191	WinVM	Software protection service scheduled successfully.	3	60642
Nov 1, 2026 @ 13:26:50.533	WinVM	Executable file dropped in folder commonly used by malware	15	92213
Nov 1, 2026 @ 13:02:50.248	WinVM	Software protection service scheduled successfully.	3	60642
Nov 1, 2026 @ 13:01:15.552	WinVM	Application Compatibility Database launched	12	92058
Nov 1, 2026 @ 12:42:10.158	WinVM	Software protection service scheduled successfully.	3	60642
[illegible]:23.175	WinVM	Software protection service scheduled successfully.	3	60642
Nov 1, 2026 @ 12:19:20.967	WinVM	Executable file dropped in folder commonly used by malware	15	92213
Nov 1, 2026 @ 12:18:20.957	WinVM	Software protection service scheduled successfully.	3	60642
Nov 1, 2026 @ 12:15:31.019	WinVM	The VSS service is shutting down due to idle timeout.	5	60702
Nov 1, 2026 @ 12:13:40.395	WinVM	Software protection service scheduled successfully.	3	60642
Nov 1, 2026 @ 12:13:06.316	WinVM	Service startup type was changed	3	61104
Nov 1, 2026 @ 12:13:01.284	WinVM	License activation (slui.exe) failed.	5	60646

c-Screenshot of the events with their description and rule level. Our alert level is (15) with rule id (92213).

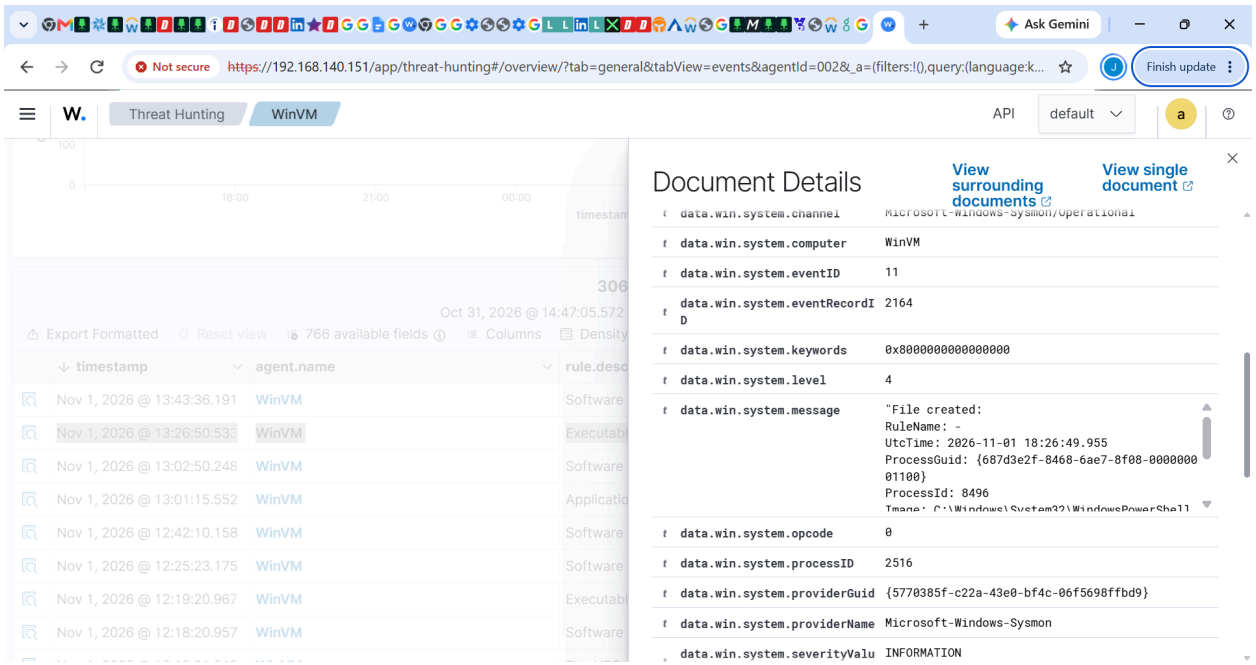


timestamp	agent.name	rule.description	rule.level	rule.id
Nov 1, 2026 @ 13:43:36.191	WinVM	Software protection service scheduled successfully.	3	60642
Nov 1, 2026 @ 13:2[illegible]	WinVM	Executable file dropped in folder commonly used by malware	15	92[illegible]
Nov 1, 2026 @ 13:02:50.248	WinVM	Software protection service scheduled successfully.	3	60642
Nov 1, 2026 @ 13:01:15.552	WinVM	Application Compatibility Database launched	12	92058
Nov 1, 2026 @ 12:42:10.158	WinVM	Software protection service scheduled successfully.	3	60642
Nov 1, 2026 @ 12:25:23.175	WinVM	Software protection service scheduled successfully.	3	60642
Nov 1, 2026 @ 12:19:20.967	WinVM	Executable file dropped in folder commonly used by malware	15	92213
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Nov 1, 2026 @ 12:15:31.019	WinVM	The VSS service is shutting down due to idle timeout.	5	60702
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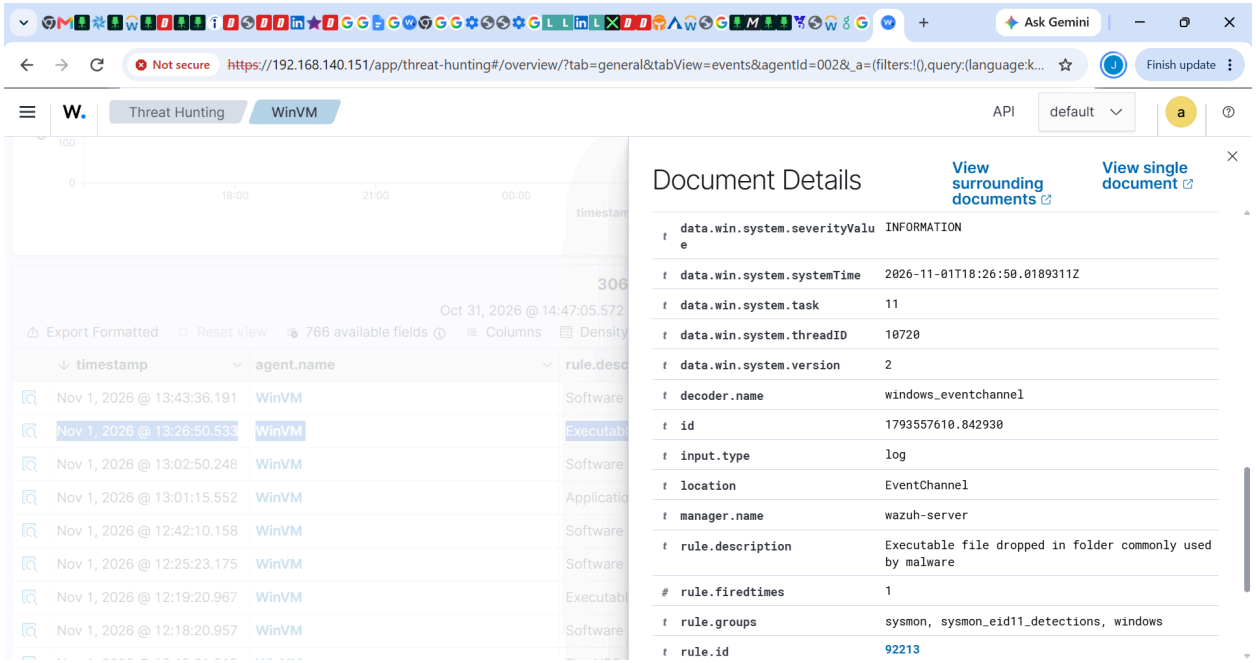
d-Screenshot of the alert highlighted with timestamp,agent name as WinVM. The criticality level is the highest on the list forcing an immediate validation process. The Screenshots in previous sections of this project validated this alert as an incident. In an enterprise setting, an investigation is immediately triggered with stakeholder's communication activated to first contain the attack with an immediate escalation to L2 with all IOCs collected through early analysis.



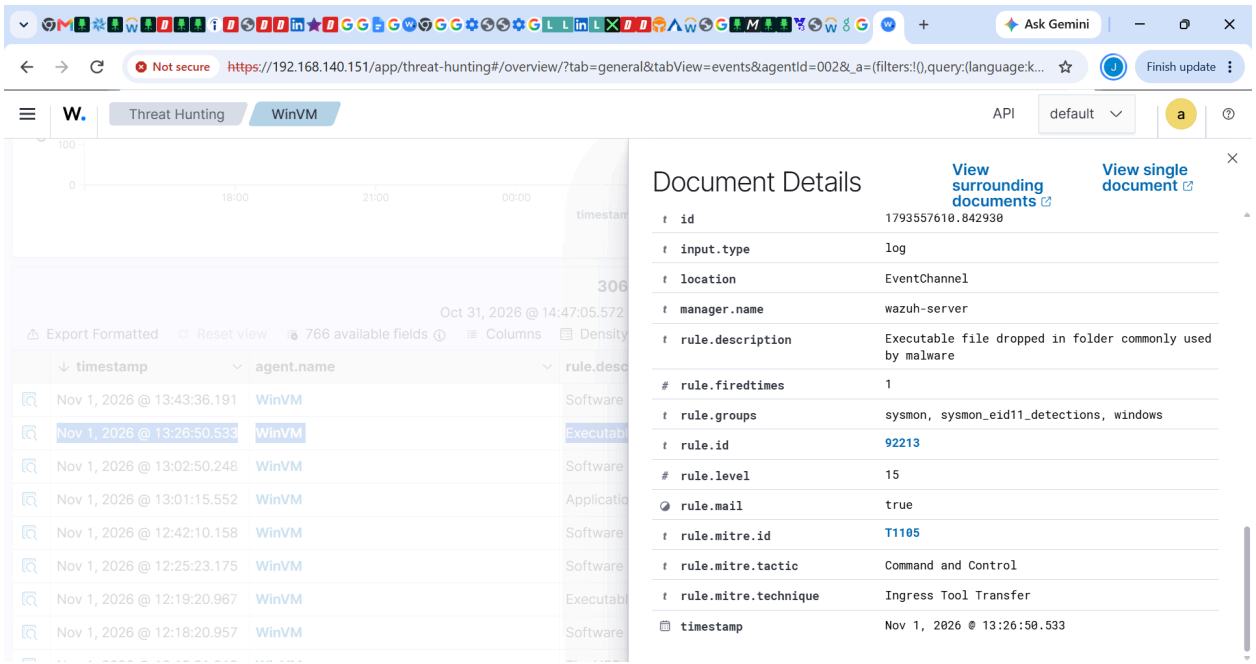
e1-Screenshot of the process path from Powershell and what the target of the action was as captured by Sysmon then forwarded to the SIEM.



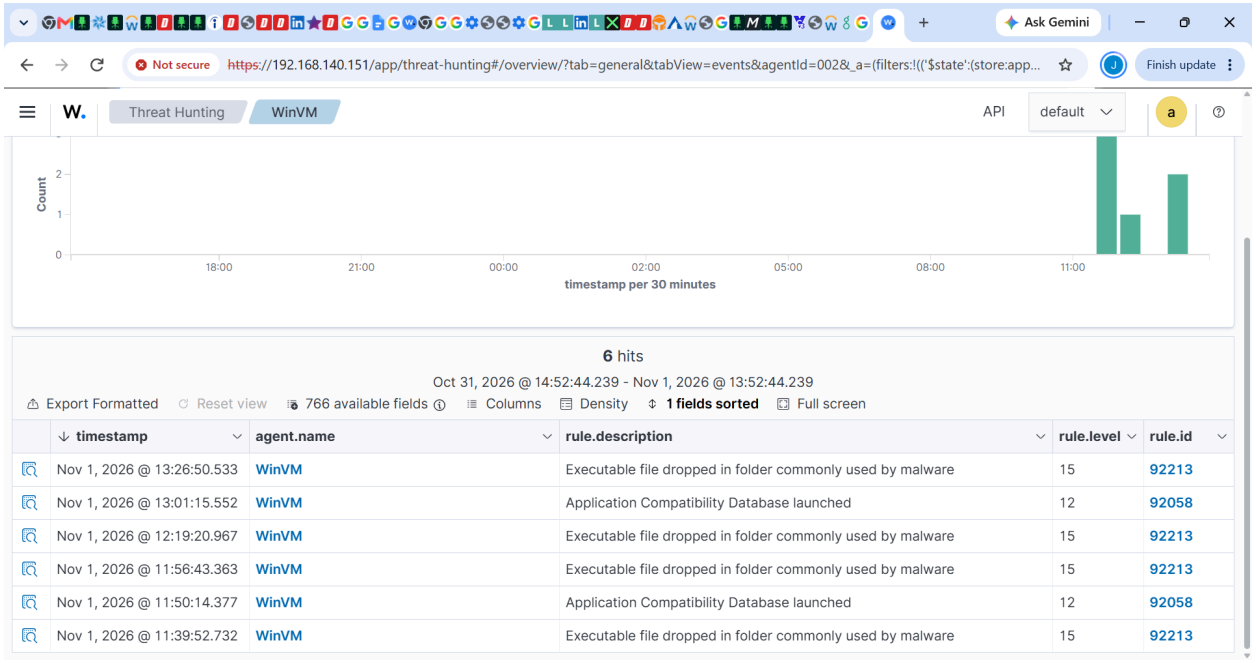
e2-Screenshot with more details from the incident table.



e3-Screenshot with table section details.

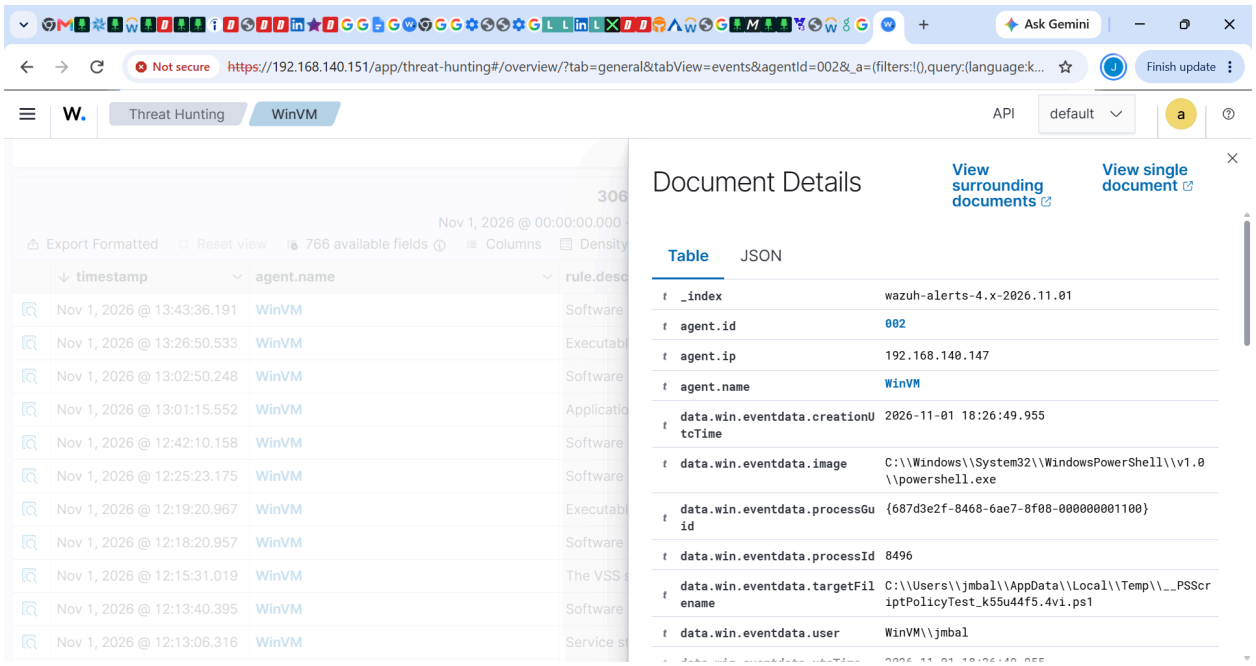


e4-Screenshot of the bottom section of the details table.

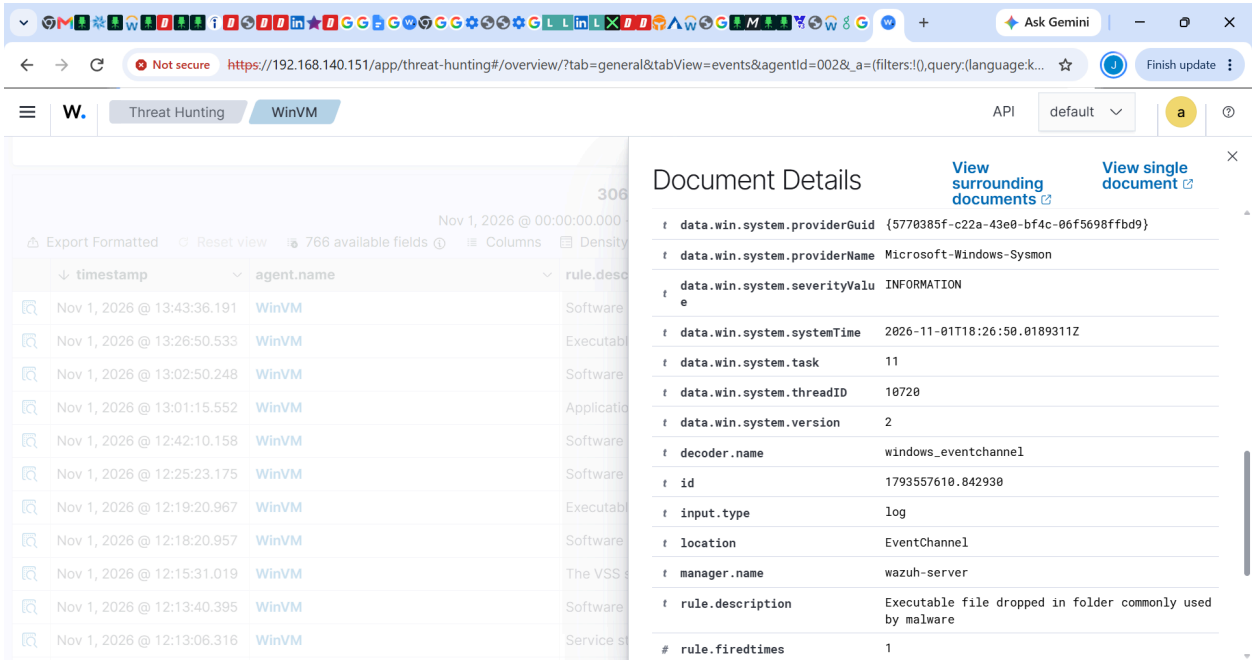


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Nov 1, 2026 @ 11:56:43.363	WinVM	Executable file dropped in folder commonly used by malware	15	92213
Nov 1, 2026 @ 11:50:14.377	WinVM	Application Compatibility Database launched	12	92058
Nov 1, 2026 @ 11:39:52.732	WinVM	Executable file dropped in folder commonly used by malware	15	92213

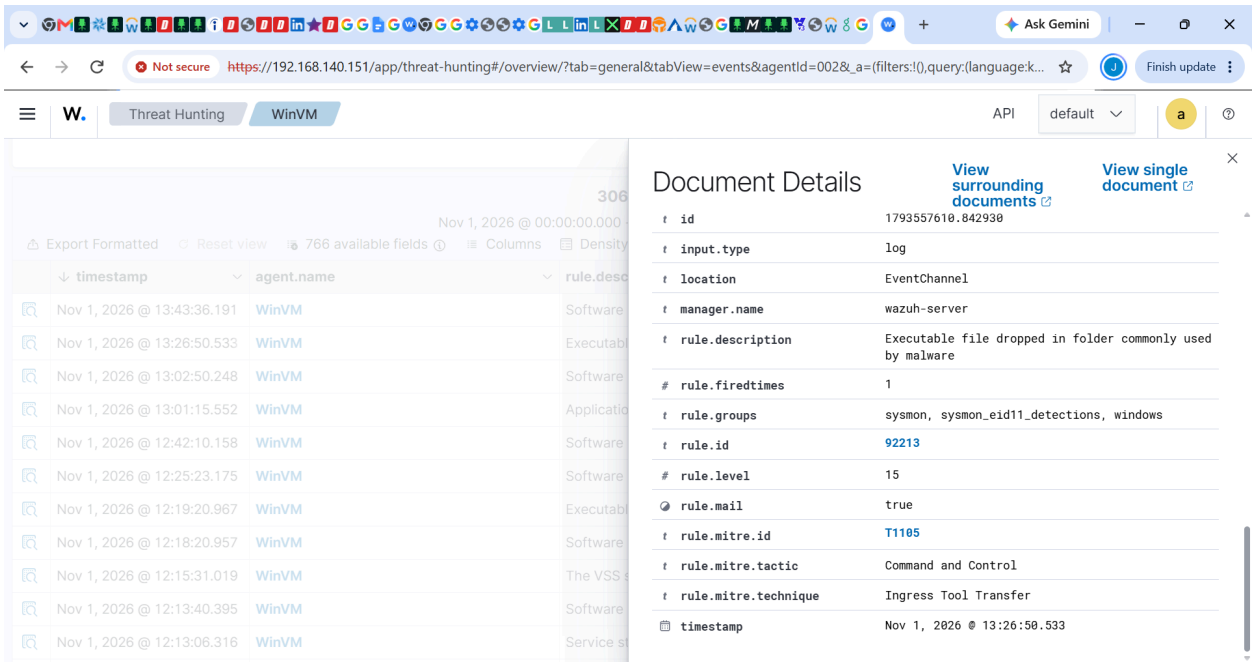
d-Screenshot of the graph captured of activity by timestamp with a spike around the time of the event.



e1-Screenshot of the policy breached in the system and the execution path.



e2-Screenshot showing the SIEM manager description of the event matching a rule in the system.



e3-Screenshot showing the SIEM and Sysmon description and rule groups matched to generate the alert.

This lab started with a Netcat protocol execution for networking connections from Ubuntu server to Kali Linux then from Ubuntu to WinVM. The following activity was to initiate a data exchange between WinVM and Ubuntu server using Netcat. After troubleshooting the connection because of earlier failure, it was established and the data exchange executed successfully. As the text sent as a test from Powershell executed in Ubuntu cleanly. The

command execution the was confirmed logged through Sysmon and the Wazuh agent on WinVM forwarded the incident data as expected to the SIEM dashboard where the analyst can track the incident point of entry, correlation telemetries, review logs and make an informed decision for validation after triage and early analysis then confidently escalate to L2 for further investigation.

Networking connections, data exchanges, SIEM aggregation and correlation were all verified throughout this lab. I used AI prompts to assist with complex troubleshooting commands and redirection during the activities.